

About:

Motivated Software Developer and AI/ML enthusiast skilled in Python, Java, JavaScript (React.js, Node.js), and SQL/NoSQL databases, with experience in web development, computer vision, and recommender systems. Strong foundations in DSA, design patterns, and system design, with a focus on writing clean, testable code and contributing effectively to the software development lifecycle (SDLC). Experienced with Linux/Unix environments, cloud platforms (AWS/GCP), distributed systems, scalable data pipelines, and machine learning workflows. Proficient in version control (Git/GitHub), CI/CD practices, software testing, debugging, and integrating AI-driven components into production-ready applications. Adaptable, detail-oriented, and committed to delivering reliable, high-quality engineering solutions.

TECHNICAL SKILLS:

Programming Languages: Python, Java, C

Web Technologies: HTML5, CSS3, JavaScript, React.js, Node.js, REST APIs

Machine Learning & AI: TensorFlow, OpenCV, CNNs, LLMs, RAG Pipelines (exposure), Data Pre-processing, Feature Engineering, Computer Vision, Recommender Systems

Data & Distributed Systems: ETL/ELT Pipelines, Data Modeling, MySQL, MongoDB

Software Engineering: OOP, Software Testing, Debugging, SDLC, Design Patterns

Tools & Frameworks: NumPy, Pandas, Anaconda, PyCharm, Docker, LangChain, Apache

Cloud & DevOps: AWS/GCP (exposure), CI/CD (basic), Git/GitHub

Operating Systems: Linux/Unix, Windows

WORK EXPERIENCE:

Software Development Engineering Intern, SK Info Techies

(May 2025 - July 2025)

- Contributed to the development and enhancement of web applications using JavaScript, React.js, and Node.js, improving UI responsiveness and user experience across multiple client-facing modules.
- Worked on database operations (MySQL/MongoDB), debugging, API integration, and version control (Git/GitHub), reducing issue resolution time by ~20% through efficient troubleshooting and clean code practices.
- Collaborated in an agile team environment, participating in code reviews, sprint tasks, and SDLC processes, which strengthened technical problem-solving and improved delivery consistency by meeting 100% of assigned deadlines.

PROJECTS:

Analytical Posture Modelling for Personalized Curriculum Optimization

- Developed a **real-time posture** classification system using **CNNs**, **OpenCV**, and upper-body keypoint detection, enabling accurate recognition of student posture variations through techniques like **Gaussian Filtering** and NMS.
- Designed a scalable analytics pipeline integrating **MySQL/SQLite**, allowing **automated tracking** of exam scores, quiz results, attention metrics, and curriculum **recommendations**, supported by structured data modeling and workflow optimization.
- Built interactive Power BI/Excel dashboards to **visualize posture-performance** correlations, enabling teachers and parents to analyze behavioral patterns, identify **weak areas**, and **support personalized** learning interventions.

Multimodal Deepfake Detection: Voice + Video

- Built a **multimodal deepfake detection** system using **LLMs**, **CNNs**, audio-video signal processing, and advanced feature extraction techniques, enabling **high-accuracy** classification of **synthetic media** across text, audio, and video modalities.
- Designed and optimized **ETL/ELT data pipelines** for **large-scale multimodal datasets**, including **preprocessing**, exploratory data analysis (**EDA**), and integration of analytics workflows to evaluate model performance and support scalable ML experimentation.

Real-Time Human Pose Detection using Deep Learning & OpenCV

- Developed a **real-time human pose detection system** using **OpenPose**, **CNNs**, and **OpenCV**, accurately identifying key body joints and classifying sitting postures (straight, hunchback, reclined).
- Implemented **heatmaps**, **Part Affinity Fields (PAFs)**, and **angle calculations** for precise posture analysis, optimizing real-time video processing with TensorFlow/Keras and OpenCV.
- Generated **analytics dashboards (Excel, Power BI, Matplotlib, Seaborn)** for performance visualization

EDUCATION:

Bachelor of Technology, Computer Science, Sharda University (GPA – 7.5/10.0)

(Sep 2022 - May 2026)

Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Artificial Intelligence, Machine Learning, Deep Learning, Pattern Recognition, Natural Language Processing, Computer Vision, Probability & Statistics, Recommender Systems

Intermediate, PCM, Lucknow Public School (GPA - 8.2/10.0)

(Mar 2021 - May 2022)

Certificate:

- Database foundation by Oracle academy (Nov 2023)
- Web Development by Udemy (Aug 2023)
- MERN Stack by Sharda University

Patent (Published):

(29 Aug 2025)

Analytical Posture Modelling for Personalized Curriculum Optimization

Patent Publication Number: 202511063027

Designed an AI-driven model using CNNs, heatmap keypoints, and PAF-based posture detection to generate automated student performance reports and personalized curriculum recommendations. Built end-to-end data pipelines and analytics modules for subject-wise evaluation.